Disclosure Regulation Stock Liquidity and Policy Diffusion Around the World

Expanded empirical draft: GDP weighting and JIBS-inspired mechanisms, 18 October 2026

Preliminary evidence conditional on documented policy cases

Abstract

We examine stock liquidity and international implementation timing across fourteen disclosure-related regulatory families. The register contains 359 documented country-family implementations matched to1,055,142 annual quote observations across 90 jurisdictions during 2000-2025. Equal-country and GDP-weighted comparisons differ materially: the IFRS spread comparison shrinks from-13.7 to-1.4 percent, and confidential country-by-country reporting reverses from-5.6 to+10.8 percent. We add a frozen illiquidity factor, trading-activity and volatility outcomes, fourteen robustness configurations, balanced event profiles and observable income splits. These comparisons remain descriptive conditional on documented cases. Following Koning et al. (2018), we separately measure export competition, relative geographic success, peer emulation and financial dependence. The fixed regression matrix yields 442 stable focal coefficients;77 have raw p below 0.05, but none survives Holm correction across the full matrix. Financial dependence is negatively associated with recorded timing and survives only the narrower four-primary-coefficient correction. Incomplete legal histories, issuer eligibility and reporting dates prevent a causal capital-market or diffusion conclusion. All unsupported and unstable tests are disclosed.

Keywords Disclosure regulation Stock liquidity Policy diffusion International capital markets

1 Introduction

Investors face an expanding set of rules governing the information that companies provide about their financial position, sustainability risks, ownership, taxes and business relationships. These rules can alter both the content of corporate information and the institutions through which it reaches markets. Their capital market consequences need not be uniform. A public financial report, a confidential submission to a tax authority and the creation of an audit oversight body change different parts of the information environment. Understanding these differences requires a common set of market outcomes together with a careful account of what each regulation requires.

We study disclosure regulation in two parts. The first examines stock liquidity around documented policy implementations across fourteen regulatory families. The second examines the timing of independent national implementations and the information that countries could have obtained from prior implementations elsewhere. Linking these questions makes it possible to distinguish the market consequences of regulation from the international processes through which regulation spreads. A policy may be attractive because connected countries have adopted it, because their markets have improved, or because its adoption conveys legitimacy. These explanations require different empirical measures.

Our outcome analysis follows the measurement logic of Krueger, Sautner, Tang and Zhong (2023). We construct annual quoted spreads, price impact and the fraction of zero-return days from

daily equity records. Their study shows how a global disclosure sample can connect average liquidity estimates to differences in implementation, enforcement and existing information provision. We extend the research setting beyond ESG reporting, while preserving the distinction between a policy's legal existence and the population of firms to which it applies. The current draft reports calendar-year market evidence. It does not reproduce their fiscal-year firm panel or their full accounting and analyst control set.

Our diffusion analysis draws on Koning, Mertens and Roosenboom (2018). We distinguish trade competition, relative foreign success, peer emulation and financial dependence. These constructs require separate networks and signals. We first retain the original portfolio-exposure exercise, then construct the export and geographic measures used in their JIBS design and partial emulation/dependence proxies. No observed network association is treated as a causal diffusion mechanism.

The expanded setting also exposes limits that can remain less visible in a single-policy study. Several regulatory families share legal acts and coordinated implementation dates. Some require public disclosure, whereas others provide information primarily to regulators. Several apply only to large multinational groups, extractive companies or digital-asset activities. Moreover, an incomplete inventory of policies cannot establish that an unrecorded country was untreated. Our analysis retains those distinctions rather than increasing the apparent sample by assigning missing histories to the control group.

The expanded tests show why weighting matters. The IFRS spread comparison is negative under equal-country and GDP weights, but its magnitude falls sharply with GDP weighting. Confidential country-by-country reporting reverses sign. Audit-reporting comparisons become negative with GDP weighting, while several other families remain positive or lack support. The JIBS-inspired models produce several raw associations, including financial dependence and recorded timing, but none survives correction across the full new test matrix.

This draft contributes a common measurement framework, a source-linked policy register and initial empirical comparisons. It also establishes which additional data are needed for a credible causal analysis. The liquidity comparisons condition on documented policy cases and use later recorded implementations as controls where support exists. The implementation-timing models condition on documented independent national cases. These estimates describe the available evidence; neither design presently identifies the unconditional effect of global regulatory adoption. The distinction is consequential for both the interpretation of results and the next stage of the paper.

2 Economic mechanisms and empirical predictions

Disclosure can improve secondary-market liquidity when it reduces differences in information among traders. Diamond and Verrecchia (1991) explain how public information can affect liquidity, participation and the cost of capital. More accessible information about firm fundamentals can lower adverse selection, attract participation and reduce transaction costs. Financial reporting, structured filing formats and public nonfinancial reports may operate through this channel. Yet additional disclosure can be costly to produce or process. Standardized requirements may provide little new information for firms that already disclose extensively, and superficially compliant reports may change neither investor beliefs nor trading. The sign and magnitude of the liquidity response are therefore empirical questions.

Our first prediction is that rules producing incremental public information for eligible firms are followed by lower quoted spreads and price impact. A lower fraction of zero-return days provides complementary evidence of improved trading activity. This prediction is conditional on actual information reaching investors. It does not imply that confidential country-by-country tax reporting or oversight-body establishment should have the same effect as a public reporting mandate. An association for those families could reflect indirect enforcement, overlapping reforms or wider changes in the market.

A second prediction concerns implementation. Requirements with broad relevant coverage, credible enforcement and clear reporting obligations should be more likely to change the information environment. Issuing authority and the possibility of explaining noncompliance can help describe those differences, as in Krueger et al. However, the concepts must be coded within each family. Environmental norms may be pertinent to ESG reporting but need not measure enforcement of beneficial ownership registration or digital-asset rules. The expanded paper should test economically relevant interactions rather than repeat identical moderator variables across unrelated regulations.

For diffusion, competition predicts that a country's regulatory choices respond to choices by countries competing for economic resources. Koning et al. measure such competition through similarity in export patterns and, alternatively, membership in the same investment-risk class. Our current portfolio network instead measures financial exposure. Holdings in a prior implementer can transmit information or reflect common investment opportunities, but they do not by themselves establish that two destinations compete. We therefore describe the resulting coefficient as an exposure association.

Learning predicts that the experience of earlier implementers influences later choices. A liquidity-based success signal connects the two parts of our study more closely than a general measure of economic growth. However, an observed decline in spreads may arise from a market cycle, changing securities in the sample or another reform. The present signal captures information that could have been visible abroad. A later design will estimate foreign policy benefits using historical data only, separate them from common shocks, and prevent information from the focal country from entering its own signal.

Emulation and coercion remain distinct predictions. Emulation concerns acceptance among relevant peers or professional communities, independently of demonstrated economic success. Coercion concerns pressure from actors with authority or resources, including explicit policy conditions. Shared-language exposure can be a candidate peer measure, while international lending can be a candidate dependence measure. Neither constitutes a verified mechanism without family-specific validation. This draft therefore does not attach emulation or coercion labels to generic geographic proximity or to unverified credit variables.

3 Policy and market data

The policy register groups capital-market-relevant rules into IFRS, ESG and CSR reporting, climate disclosure, ISSB-based reporting, XBRL filing, governance disclosure, beneficial ownership, confidential country-by-country reporting, public country-by-country reporting, expanded audit reporting, audit oversight, extractive-payment disclosure, supply-chain rules and digital-asset regulation. Table 1 separates dated implementations from published proxies, future schedules and coordinated reference records. These categories describe evidence and event type. They do not imply that all events have the same treatment intensity or economic content.

The accompanying regulation archive serializes all 441 event records into a master JSON register and one JSON file per event, with fourteen family indices. Each record preserves its original CSV fields, source URL, locator, date distinctions and verification status. These records include implementations, amendments, published proxies and other events; they are not a count of unique statutes. The separate inherited ESG register contains 51 document codings. They may overlap the expanded events and do not add verified implementation cases to the estimation sample.

The source archive contains 258 unique preserved files or responses linked to 151 recorded source URLs after removing fragments. SHA-256 hashes identify saved objects, while retrieval logs retain failed requests and relocated official documents. Of the 441 events, 373 have an original or linked official source with readable text and 39 published-proxy records have saved author repository files. Eleven records have only a saved text derivative or partial excerpt; eighteen lack readable source content, including metadata-only captures. All 51 inherited ESG document records have saved files and extracted text. Saving a response or reading an author workbook does not independently verify a legal date or establish a complete global policy history.

The extraction pass examined 4,593 PDF pages. Tesseract was run on 206 page images when native text was sparse or corrupted, including image-only sources. Original files, page-level text, requested OCR languages, word-confidence diagnostics and searchable derivatives are retained separately. An image-ink heuristic flags 33 likely blank images; 51 nonblank or low-confidence OCR pages require review. Extraction and archiving are reproducibility steps, rather than legal or translation verification. Table A4 reports source availability by family. No treatment coding or empirical estimate is revised by this archival pass.

The register is a collection of documented implementations rather than an exhaustive history of nonadoption. Every family retains an incomplete-census flag. The absence of a record consequently does not identify an untreated country. The first empirical comparisons use countries with dated records in the same family, retaining later cases as potential controls only before their recorded implementation. Earlier unrecorded obligations cannot be ruled out by that rule alone. We accordingly refer to first recorded implementation rather than verified first adoption throughout the empirical results.

Date meaning is equally important. The register contains reporting-period starts and ends, compliance deadlines, legal application dates and institutional establishment dates. A first fiscal year is not a first public filing date. The expanded study ultimately requires separate announcement, enactment, application, reporting and publication fields. The current calculations retain the register's event year and examine liquidity in calendar year g+1, relative to calendar year g-1. This convention is transparent and comparable across observed cases, but it is an approximation to information availability.

Two differences within the existing ESG records illustrate the importance of scope. The register contains an official Indian implementation record for 2012, while the supplied Krueger manuscript uses 2015 to represent the extension from the largest one hundred to the largest five hundred listed firms. The two dates answer different coverage questions. The inventory also includes literature-based ESG proxies that do not correspond exactly to the thirty-five-country estimation sample in the supplied manuscript. We keep those proxies out of the dated-implementation analysis. Their presence is not a substitute for a new legal audit.

The daily equity source consists of five Datastream Parquet collections. We use the complete calendar years 2000 through 2025 and exclude partial 2026 observations. The records provide quote identifiers, listing and company-country fields, prices, bid and ask quotes, local-currency total returns, trading volume in shares, and market capitalization. The raw collection remains on the external drive; the project contains links to it. All derived annual files and screening audits are stored separately from the raw data.

We select equity securities flagged as major securities and primary country quotes, require the quote identifier to equal the primary quote identifier, and match trading country to company country under the source's country-code type. We remove observations after available delisting dates and apply a limited name screen for funds, warrants, depositary receipts and preferred securities. This is not the complete historical-constituent and industry screen in Krueger et al. The sample retains inactive securities present in the source, but we cannot establish that the extract exhaustively covers every historical listing.

The raw files can contain several historical ISIN mappings for the same quote and trading date. Identical economic observations are collapsed using the quote identifier, date, country, price, quote, return, volume and market-capitalization fields. We do not count additional static identifiers as additional trading observations. If economic observations conflict at the same quote-date key, the key is excluded rather than selecting an arbitrary value. The yearly audit reports the remaining

conflicts and checks recorded returns against changes in the total-return index. A boundary-window probe finds differences concentrated in Monday observations from several markets; unavailable local trading dates are a possible explanation. We retain vendor returns and leave full trading-calendar reconciliation open, rather than overwrite them with returns across observed-date gaps.

Daily prices below USD 0.10 are excluded. Returns above 200 percent and abnormal adjacent return reversals are removed following the thresholds described in Krueger et al. Daily quoted spreads require nonnegative bid and ask prices and an ask at least as large as the bid. Dollar trading value is the closing USD price times volume in shares; observations below USD 100 do not enter price impact. We infer active country trading dates from at least five selected securities with positive volume, allowing the trading week to differ across countries.

Annual spread is the median daily relative spread. Annual price impact is the median absolute decimal total return divided by USD trading value, rescaled to match the units described in the reference paper. Spread and impact are winsorized at the first and ninety-ninth percentiles within each year and enter the comparisons in logarithms. We require at least one hundred twenty valid daily observations for each measure. Price impact and zero-return fractions additionally require variation in closing prices.

The zero-return outcome divides zero-return days by observed valid-return days on the inferred country trading calendar. It does not reproduce a denominator from a separately audited exchange calendar. An audit of the separate LSEG ESG files locates 175,506 fiscal year-end dates for 15,592 Datastream codes. These cover a subset and still require a validated historical link to the quote panel. The inspected ESG field definitions do not provide the full assets, profitability, debt, analyst and guidance control set. These differences are recorded in the analysis JSON and prevent an exact replication claim. Table 2 reports coverage and distributions for the measures that are actually constructed.

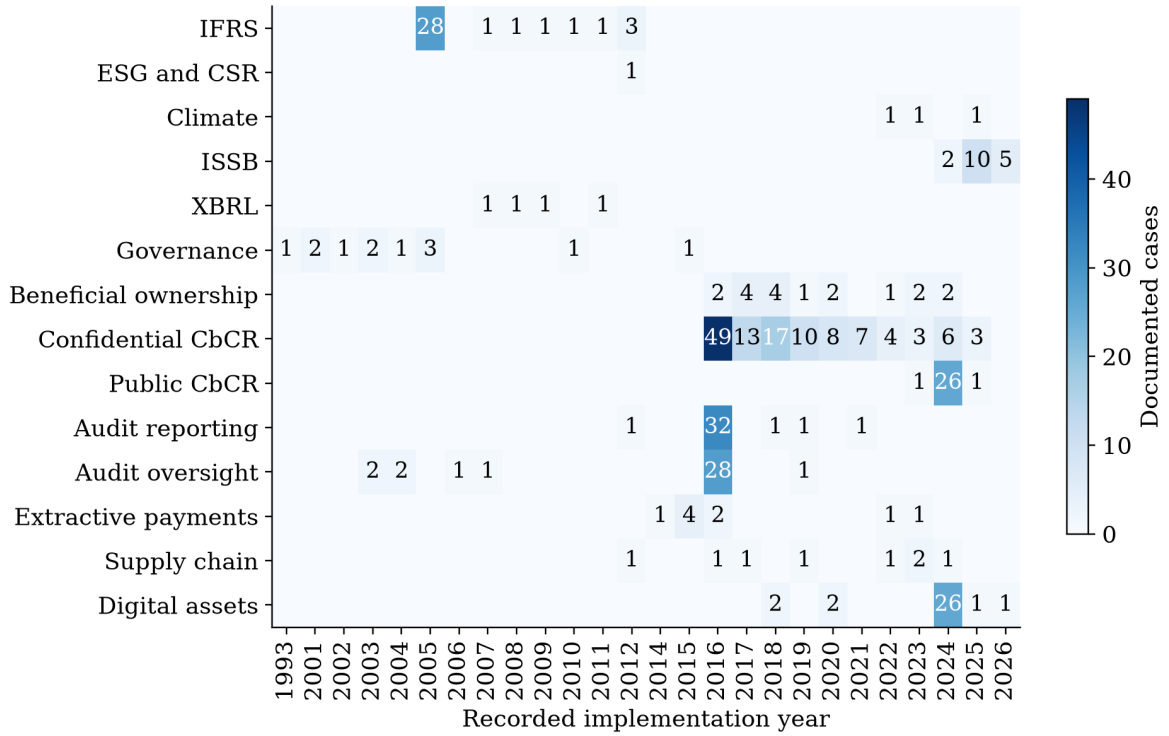
The screened annual files contain 1,055,142 security-years for 99,100 distinct quote identifiers in 90 jurisdictions. Log spreads are available in 662,316 security-years, log price impact in 755,877, and the zero-return fraction in 795,229. The common three-outcome sample contains 636,960 observations. These counts describe the market-data build and are not the number of firms treated by a policy. Table 2 shows the corresponding distributions, while Figure 2 makes the time variation in coverage visible. The country comparisons further require thirty securities with a valid observation in both comparison years.

Table 1 Policy coverage and documented implementation cases

Policy family	Dated cases	Independent cases	Proxy records	Future schedules
IFRS	36	11	0	0
ESG and CSR	1	1	39	0
Climate	3	3	0	0
ISSB	17	17	0	6
XBRL	4	4	0	0
Governance	12	12	0	0
Beneficial ownership	18	12	0	0
Confidential CbCR	120	93	0	0
Public CbCR	28	1	0	0
Audit reporting	36	9	0	0
Audit oversight	35	7	0	0
Extractive payments	9	4	0	0
Supply chain	8	8	0	1
Digital assets	32	4	0	0

Dated cases are country-family implementations in the current source register, not a complete global adoption census. Independent cases follow the recorded national-choice flag. Proxy and future-schedule records are excluded from the dated-case estimates. Coordinated reference dates and other events are separately retained in the source CSV. All fourteen families have incomplete-census flags.

Figure 1 Timing of documented implementations



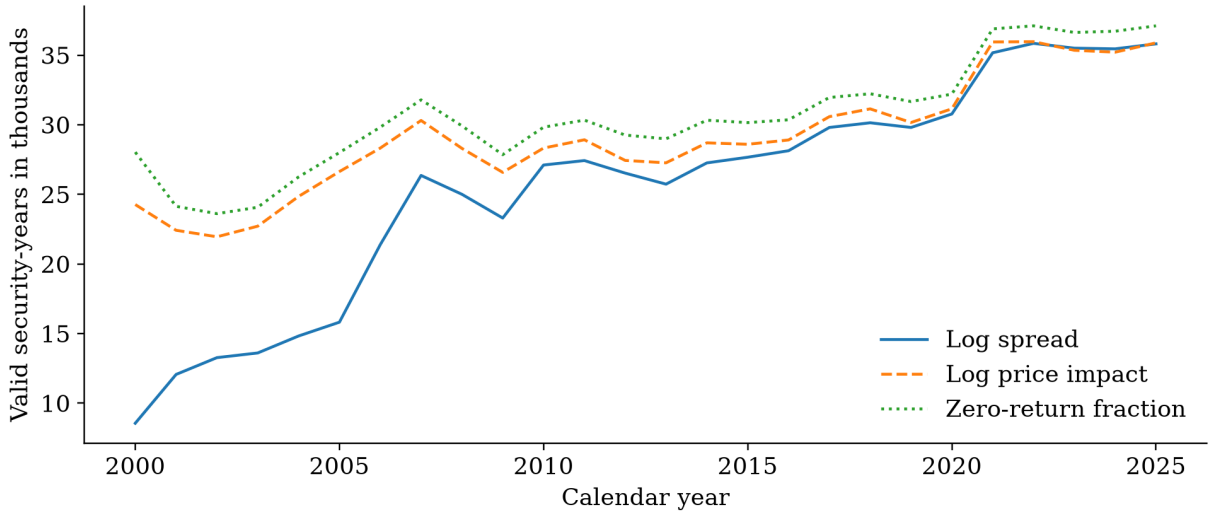
Cells show recorded country-family case counts. Common legal acts can generate many same-year national records. Counts are descriptive; they do not measure independent treatment decisions or the global adoption rate.

Table 2 Security year outcome distributions

Outcome	N	Mean	SD	Median	P95
spread	662,316	0.016	0.024	0.006	0.061
impact	763,729	123.846	404.371	4.754	657.203
zero fraction	795,229	0.119	0.153	0.064	0.400
log spread	662,316	-5.074	1.436	-5.047	-2.803
log impact	755,877	1.516	3.079	1.603	6.500

Distributions are outcome-specific after annual screens. Spread and price impact are winsorized within year at 1 and 99 percent. Price impact uses decimal returns and USD trading value with the reference-paper rescaling. Zero fraction uses observed valid returns, not an independently audited exchange-calendar denominator. Raw quote identifiers are the unit; no complete historical issuer consolidation is claimed.

Figure 2 Availability of annual liquidity outcomes



Counts follow the outcome-specific minimum of 120 daily observations and other stated filters. Changes in coverage need not reflect regulatory effects.

4 Liquidity around documented implementations

We begin with annual quoted spreads, price impact and zero-return fractions following the measurement logic of Krueger et al. (2023). Table 3 retains the initial equal-country comparisons. The follow-up fixes the same supporting quote identifiers within each country between g-1 and g+1, requires thirty paired quotes per country and at least three later-case controls per cohort, and compares equal-country results with alternative country weights. Missing histories do not supply never-treated observations.

GDP weights answer a different aggregation question from equal-country weights. We use current USD GDP, constant 2015 USD GDP and constant 2021 international-dollar PPP GDP measured at g-1. We also use baseline sampled-quote market capitalization and balanced-quote counts. We normalize weights separately in each cohort arm and aggregate cohorts by treated weight mass. Equal-country results on exactly the same eligible country/cohort cells distinguish lost GDP coverage from weighting changes. Economic size does not resolve endogenous timing, firm eligibility or concurrent reforms.

The IFRS log-spread comparison changes from -0.147 (-13.7%) to -0.014 (-1.4%) under constant-GDP weights. Confidential CbCR changes from -0.057 (-5.6%) to 0.103 (+10.8%). These reversals and attenuations demonstrate sensitivity to economic composition; they cannot establish a common policy benefit.

Audit reporting changes from a positive equal-country spread contrast to a negative GDP-weighted contrast. Its GDP-weighted cluster-deletion range remains below zero in the baseline. This is a descriptive candidate for further investigation, not formal evidence of a policy effect: coordinated cases, sample composition and unobserved overlapping changes can still drive

it. Figure 3 and Table 4 show all families, including unsupported comparisons, rather than selecting the negative estimates.

We add a principal-component illiquidity index, log traded value relative to market capitalization and daily-return volatility. PC1 is estimated once on 57,097 complete observations during 2000-2004, with frozen standardization and loadings applied to all years. It explains 67.8 percent of reference covariance and increases with illiquidity. Some policies precede this calibration window, so it is a retrospective measurement reference. Table 5 reports the added outcomes. Traded value divided by median market cap is an activity ratio rather than exact share turnover.

The frozen follow-up plan specifies fourteen configurations: baseline, quote-cap weighting, median quote changes, common-three-outcome support,200-day availability, removal of the lowest pre-cap quintile, raw proxies,5/95 winsorization, independent treated cases, independent cases in both arms, known-other-family event exclusions, crisis-window exclusions, pretrend extrapolation and average post-years 1-3. We disclose all 3,330 planned weighting/robustness cells:2,314 are supported and 1,016 are unavailable or inapplicable. These are related analyses, not 3,330 independent experiments. No filters are combined opportunistically.

Figure 4 uses the same quotes, countries and cohorts observed at every year from g-3 through g+3, with later-case controls implementing after g+3 and g-1 as reference. The balanced profiles repair the varying-support limitation of AppendixA 1, but descriptive pre-period patterns do not establish parallel trends. Figure 5 and TableA 5 disclose the robustness grid. An additional, explicitly exploratory income split applies each pre-year median GDPpc threshold to both comparison arms; its full 180-cell table includes unsupported groups.

The separate published-ESG benchmark retains the supplied 35 treated and 30 never-coded countries only during 2002-2020. Its equal-country log-spread comparison is-0.016 and its constant-GDP comparison is-0.052. Current-USD GDP weighting produces a different sign, illustrating that vintage/currency composition and eligible support matter even in the reference sample. This is a calendar benchmark using published dates, not a fiscal-year replication of Krueger et al. Formal liquidity p-values are withheld; assignment-cluster deletion ranges remain sensitivity diagnostics, not confidence intervals.

$$\widehat{\Delta}_{k,g,h} = \overline{(Y_{g+h} - Y_{g-1})}_{G_k = g} - \overline{(Y_{g+h} - Y_{g-1})}_{G_k > \max(g, g+h)}$$

Equation 1. Country changes first balance securities within each pair of years; bars then denote equal-country means. Comparisons require three supported control countries. The primary horizon is h=1. Aggregation weights each supported cohort by its contributing treated-country count.

Table 3 Liquidity comparisons conditional on documented cases

Family	Outcome	Change	Treated countries	Cohorts	Policy clusters	Cluster deletion range
IFRS	Log spread	-0.147	20	5	12	-0.293 to 0.011
IFRS	Log impact	-0.167	21	5	12	-0.297 to 0.128
IFRS	Zero fraction	-0.028	22	5	12	-0.032 to -0.023
ESG and CSR	Log spread	Not estimated	-	-	-	-
ESG and CSR	Log impact	Not estimated	-	-	-	-
ESG and CSR	Zero fraction	Not estimated	-	-	-	-
Climate	Log spread	Not estimated	-	-	-	-
Climate	Log impact	Not estimated	-	-	-	-
Climate	Zero fraction	Not estimated	-	-	-	-
ISSB	Log spread	0.061	1	1	6	-0.024 to 0.147
ISSB	Log impact	-0.001	1	1	6	-0.151 to 0.075
ISSB	Zero fraction	0.011	1	1	6	0.009 to 0.014
XBRL	Log spread	Not estimated	-	-	-	-
XBRL	Log impact	Not estimated	-	-	-	-
XBRL	Zero fraction	Not estimated	-	-	-	-
Governance	Log spread	0.030	5	3	10	-0.030 to 0.260
Governance	Log impact	0.162	5	3	11	0.034 to 0.341
Governance	Zero fraction	0.019	5	3	11	0.014 to 0.028
Beneficial ownership	Log spread	0.032	11	5	10	-0.007 to 0.091
Beneficial ownership	Log impact	-0.052	11	5	10	-0.143 to 0.052
Beneficial ownership	Zero fraction	0.008	11	5	11	0.004 to 0.013
Confidential CbCR	Log spread	-0.057	42	4	33	-0.101 to -0.028
Confidential CbCR	Log impact	-0.166	42	4	33	-0.233 to -0.103
Confidential CbCR	Zero fraction	-0.002	43	4	35	-0.008 to 0.003
Public CbCR	Log spread	Not estimated	-	-	-	-
Public CbCR	Log impact	Not estimated	-	-	-	-
Public CbCR	Zero fraction	Not estimated	-	-	-	-
Audit reporting	Log spread	0.025	20	2	10	-0.112 to 0.038
Audit reporting	Log impact	0.199	20	2	10	0.183 to 0.488
Audit reporting	Zero fraction	0.000	20	2	10	-0.012 to 0.005
Audit oversight	Log spread	0.029	4	3	6	-0.100 to 0.103
Audit oversight	Log impact	0.014	6	4	8	-0.195 to 0.126

Family	Outcome	Change	Treated countries	Cohorts	Policy clusters	Cluster deletion range
Audit oversight	Zero fraction	-0.002	6	4	8	-0.016 to 0.002
Extractive payments	Log spread	0.210	1	1	4	0.219 to 0.229
Extractive payments	Log impact	0.499	1	1	4	0.505 to 0.522
Extractive payments	Zero fraction	-0.011	1	1	4	-0.011 to -0.009
Supply chain	Log spread	0.066	4	4	8	0.028 to 0.095
Supply chain	Log impact	-0.143	4	4	8	-0.225 to -0.039
Supply chain	Zero fraction	-0.013	4	4	8	-0.018 to -0.009
Digital assets	Log spread	0.016	2	2	4	-0.002 to 0.045
Digital assets	Log impact	0.260	2	2	4	0.015 to 0.526
Digital assets	Zero fraction	-0.017	2	2	4	-0.024 to -0.011

Outcome year g+1 versus g-1, balancing securities within each country and comparison. Later recorded cases must implement after the outcome year. Each country requires at least 30 balanced securities and each cohort at least three control countries. Cohorts are weighted by contributing treated-country counts. Estimates are log points for spread/impact and fractions for zero returns. The range deletes one recorded assignment cluster at a time and recalculates supported comparisons; it is a sensitivity range, not a confidence interval. Cohort composition can change across deletions. Formal liquidity-effect inference is withheld. Detailed country cells and deletion support are in the CSV release.

Table 4 Log spread comparisons under alternative country weights

Family	Equal	GDP-matched equal	Constant GDP	Market cap	GDP effective N	Largest GDP weight %
Audit oversight	0.029	0.029	-0.109	-0.077	1.5	80.1
Audit reporting	0.025	0.025	-0.058	-0.066	9.3	19.3
Beneficial ownership	0.032	0.032	0.067	0.100	7.1	21.4
Confidential CbCR	-0.057	-0.057	0.103	0.115	8.6	26.9
Public CbCR	-	-	-	-	-	-
Climate	-	-	-	-	-	-
Digital assets	0.016	0.016	-0.030	-0.027	2.0	54.3
ESG and CSR	-	-	-	-	-	-
Extractive payments	0.210	0.210	0.159	0.157	1.0	100.0
Governance	0.030	0.030	0.226	0.254	2.4	59.2
IFRS	-0.147	-0.147	-0.014	-0.181	9.9	17.2
ISSB	0.061	0.104	-0.155	-0.096	1.0	100.0
Supply chain	0.066	0.066	0.004	0.028	1.9	70.7
XBRL	-	-	-	-	-	-
Published ESG benchmark	-0.016	-0.012	-0.052	-0.104	9.8	22.9

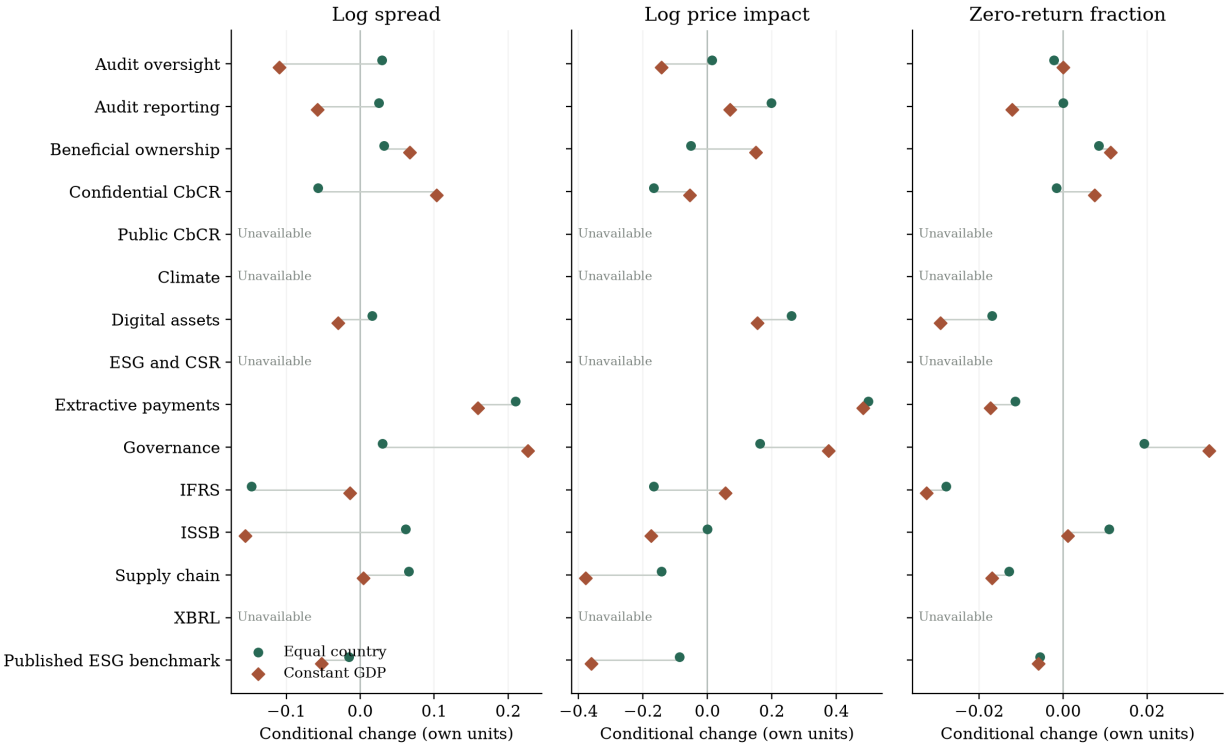
Year g+1 versus g-1. GDP and market-cap weights are fixed at g-1, normalized in each cohort arm; cohorts receive treated weight mass. GDP-matched equal uses the exact country/cohort cells with observed GDP but retains equal-country aggregation. Effective N=1/sum(squared treated-country weight shares); it is not a sampling degrees-of-freedom count. Market cap sums sampled quotes, not the full national market. A dash is unsupported. The published ESG benchmark remains separate. All six weighting schemes and three original outcomes are released in CSV.

Table 5 Additional market outcomes

Family	PC1 equal	PC1 GDP	Trading/cap equal	Trading/cap GDP	Volatility equal	Volatility GDP
Audit oversight	0.102	-0.125	-0.067	0.079	-0.003	-0.004
Audit reporting	0.057	-0.094	-0.207	-0.139	-0.000	-0.000
Beneficial ownership	0.032	0.120	-0.101	-0.191	-0.000	-0.000
Confidential CbCR	-0.083	0.086	-0.013	-0.191	-0.001	-0.003
Public CbCR	-	-	-	-	-	-
Climate	-	-	-	-	-	-
Digital assets	0.007	-0.107	-0.144	-0.109	0.001	-0.000
ESG and CSR	-	-	-	-	-	-
Extractive payments	0.220	0.161	-0.107	-0.077	0.002	0.001
Governance	0.085	0.298	-0.113	-0.105	0.002	0.005
IFRS	-0.289	-0.107	0.221	0.217	0.003	0.002
ISSB	0.073	-0.129	-0.506	-0.538	-0.009	-0.009
Supply chain	-0.024	-0.138	0.087	0.100	0.001	-0.002
XBRL	-	-	-	-	-	-
Published ESG benchmark	-0.061	-0.153	-0.022	0.023	-0.001	-0.005

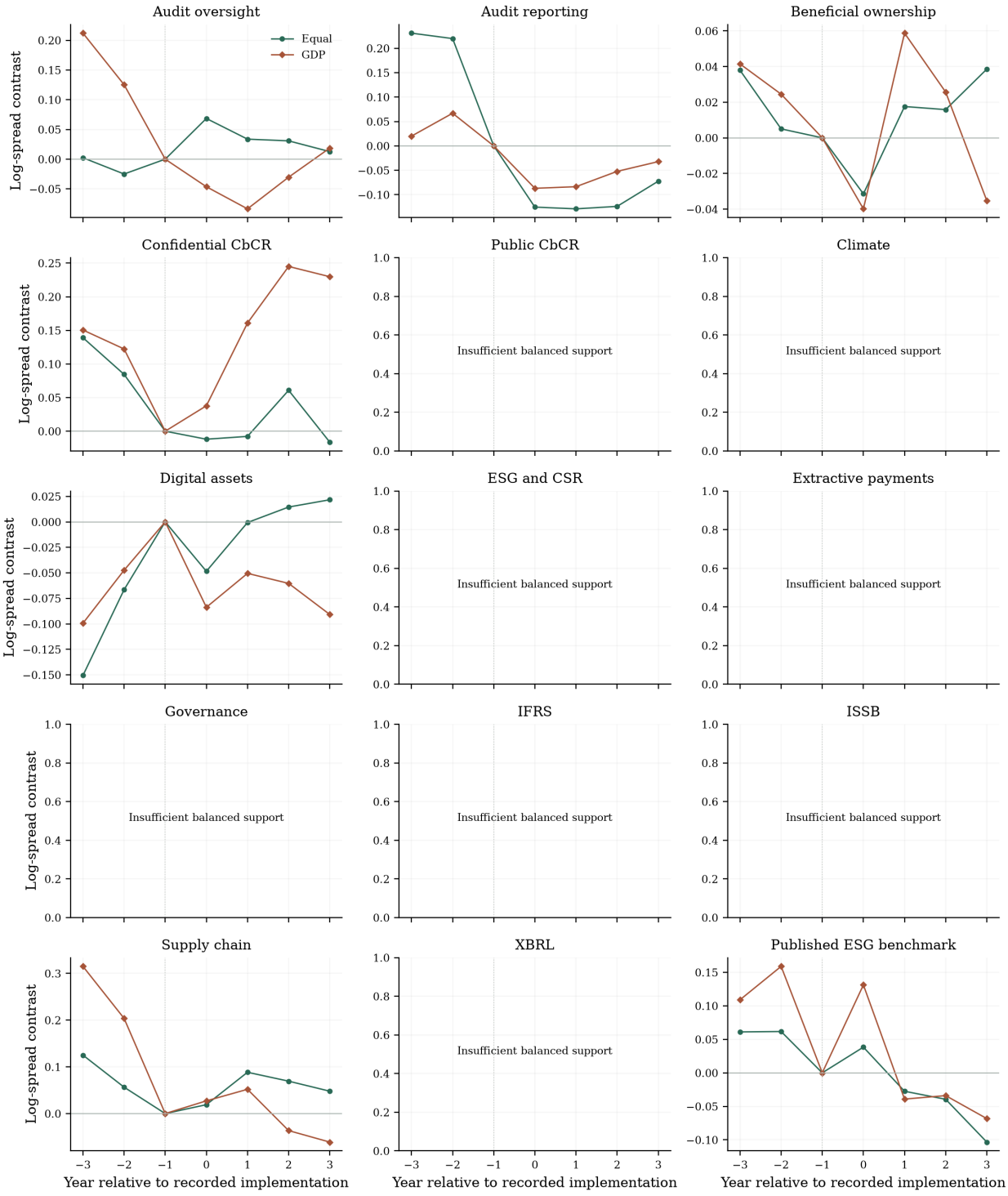
Each cell applies the baseline balanced-quote comparison. PC1 uses the original three proxies standardized on 57,097 complete 2000-2004 observations. Loadings are spread 0.651, impact 0.579 and zero returns 0.491; variance explained is 67.8 percent. Higher PC1 is more illiquid. This is a frozen retrospective measurement reference, not universally pretreatment. Trading/cap is log annual dollar traded value / annual median USD market cap; it is not exact share turnover. Volatility is daily-return SD, decimal units, with 120 valid days and price variation.

Figure 3 Equal-country and GDP-weighted liquidity comparisons



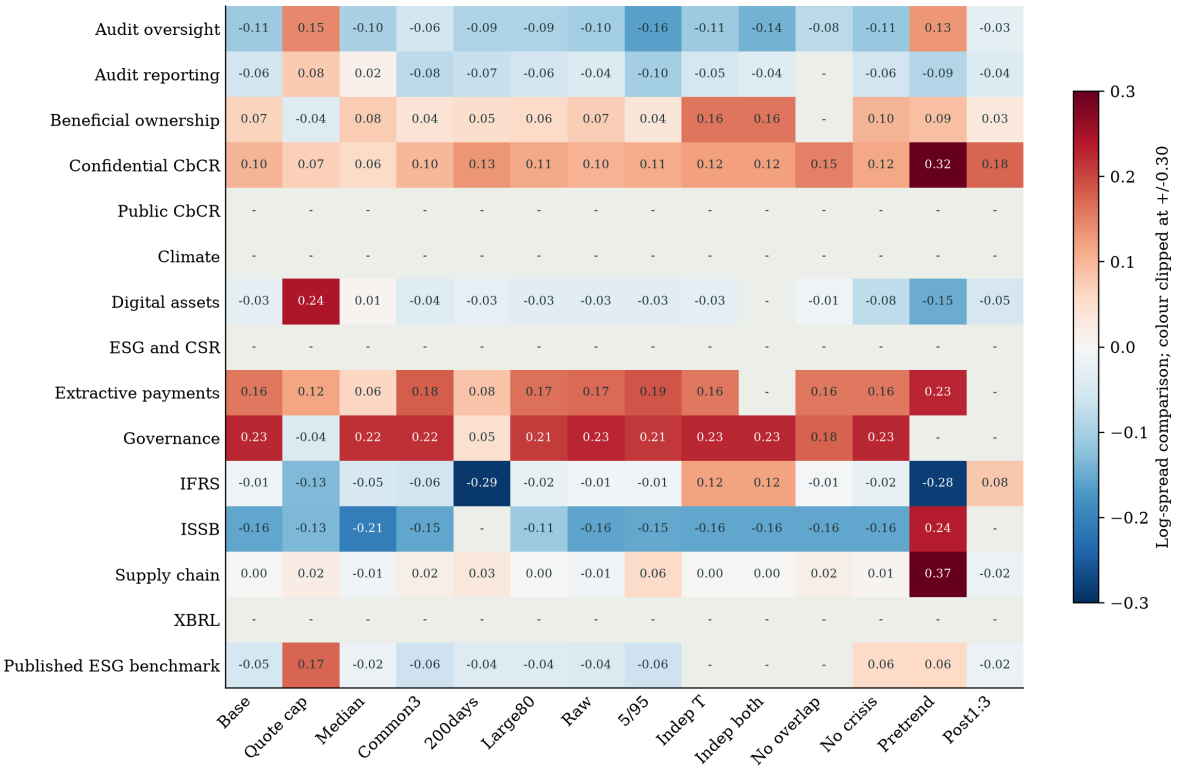
All fourteen families and the separate published ESG benchmark. Points show g+1 versus g-1 comparisons in outcome-specific units. Lines connect weighting alternatives; they are not uncertainty intervals. Matched-support equal comparisons and all six weights are downloadable.

Figure 4 Balanced log-spread profiles



Same quotes and country/cohort cells across g-3 through g+3, relative to g-1, controls recorded after g+3. Missing panels fail the fixed support requirements. Equal and constant-GDP lines are descriptive profiles, with no causal confidence bands.

Figure 5 Complete GDP-weighted log-spread robustness grid



Values are conditional comparisons. Colour scale clipped at+/-0.30; annotations retain values. Columns: baseline; quote cap; median; common three outcomes;200 days; exclude bottom 20 percent cap; raw;5/95; independent treated; independent both arms; no known overlapping family event; no crisis; pretrend adjustment; post-years 1-3. Dash means unsupported/inapplicable.

5 International implementation timing

The initial portfolio-exposure exercise is preserved in Tables 6-7. Its two stable exposure-only fits are imprecise; learning fits were unstable. Portfolio exposure is not the export-competition construct in Koning et al. (2018). The expansion therefore builds separate mechanism measures and reports their relation to the supplied JIBS design in Table 8.

For competition we select the five most correlated export-destination patterns using raw COW trade flows and at least twenty jointly observed destinations. Missing/negative flows are not converted to zero. Fixed 2000 peers are available from 2001; fixed 2013 peers enter only from 2014; annual lagged flows stop at2014 and therefore focal 2015. We calculate adoption shares only among documented dated peers, require at least three of five (four in a stricter sensitivity), and include unknown-peer coverage. The global adoption fraction remains bounded by the separately retained unknown-neighbour share.

For learning, the ten nearest geographic neighbours are selected independently of policy status. Earlier cases have recorded implementation by t-2; comparison cases implement after t-1. We compare median GDP growth, and GDPpc growth in an alternative, observed at t-1, requiring at least one measured earlier and later case and 80 percent signal coverage in each group.

Liquidity-change versions use the same temporal rule. A further foreign-success proxy uses balanced g-1 to g+1 foreign contrasts available by t-1 and excludes the focal country and its assignment cluster from foreign controls. These remain observed conditional comparisons, not causal estimates of foreign policy success.

For emulation we use separate language, legal-origin and generic IGO peer measures. These do not reproduce the Berry cultural/administrative composite or policy-specific IFAC professional advocacy. Financial dependence uses lagged IMF credit including SDR allocations and net aid relative to GDP, separately and combined. Missing higher-income reporting is retained as missing; no observed loan condition is inferred from these financial stocks and flows.

We fit first recorded implementation timing among documented independent cases and a repeated absorbing recorded-status index for comparison with JIBS. The latter is an index built from recorded implementation dates, not an audited full legal history. Single-mechanism logits include family-year strata and peer-coverage terms. Controlled variants add lagged log GDPpc, GDP growth and trade/GDP. We also fit combined four-mechanism specifications and same-sample linear probability models. Constant-outcome strata are excluded; minimum support is50 rows,15 countries,10 ones and 10 zeros. Designs must be full rank and fits must converge without warnings or invalid standard errors.

The fixed matrix contains 1,190 designs and 2,992 focal-coefficient cells including LPM counterparts and unsupported fits. There are 442 stable focal coefficients across 412 models. Of these,77 have raw p below 0.05, but none passes Holm correction across all 442 estimable coefficients. These counts do not describe independent hypotheses. Failed and unsupported specifications, exact samples and model diagnostics are retained.

In the pooled timing baseline, the export-competitor coefficient is0.395 (raw p0.144), relative-GDP-growth learning is-0.409 (p0.558), language-peer exposure is0.096 (p0.749), and combined credit/aid dependence is-0.696 (p0.000226). The latter passes the separately identified four-primary-coefficient Holm family (p0.000905), but its all-test Holm p is0.0995 and its sign is negative. It cannot establish policy-specific coercion. The baseline samples range from 72 rows/31 countries for learning to372 rows/67 countries for language exposure. Sample changes across measures and controls constrain comparison.

Table 9 and Figure 6 contrast timing and recorded-status associations. The pooled benchmarked-liquidity-success status coefficient is negative, with raw p0.0041 and all-test Holm p1.000; its first-event timing counterpart lacks the fixed support. This distinction matters because accumulated status and the timing of an independent national decision ask different questions. The present evidence does not establish that foreign liquidity benefits caused regulatory diffusion. It identifies which mechanism measures can now be formed and where legal-history and source coverage still prevent a credible test.

$$\Pr(E_{c,k,t} = 1 \mid \mathcal{D}_k) = \Lambda(\alpha_{k,t} + \beta X_{c,k,t-1} + \gamma L_{c,k,t-1} + \eta U_{c,k,t-1} + \theta' Z_{c,t-1})$$

Equation 2. D is the documented independent-case set. Lambda is the logistic link; X is prior recorded portfolio exposure, L observed foreign liquidity improvement, U the unclassified portfolio share, and Z observed domestic market controls. Family-year intercepts are denoted alpha. The exposure-only model omits L.

Table 6 Implementation timing sample support

Family	Risk rows	Observed inputs	Identifying rows	Events	Exposure model
IFRS	96	0	0	0	Not estimated
ESG and CSR	12	0	0	0	Not estimated
Climate	70	0	0	0	Not estimated
ISSB	298	0	0	0	Not estimated
XBRL	35	0	0	0	Not estimated
Governance	54	0	0	0	Not estimated
Beneficial ownership	242	29	0	0	Not estimated
Confidential CbCR	1752	448	61	31	Fitted
Public CbCR	24	0	0	0	Not estimated
Audit reporting	150	61	0	0	Not estimated
Audit oversight	46	16	9	3	Not estimated
Extractive payments	74	3	0	0	Not estimated
Supply chain	156	5	0	0	Not estimated
Digital assets	76	0	0	0	Not estimated

Risk rows belong to documented independent cases only. Observed-input rows meet the 80 percent portfolio classification rule and observed market-control requirements. Identifying rows also remove constant-outcome family-year strata. Estimates require at least 50 rows, 15 countries and 10 events. The learning model has its own signal-coverage attrition in the accompanying CSV. A non-estimated family can have zero rows because all dated focal cases are coordinated.

Table 7 Conditional implementation timing estimates

Sample	Model	Term	Log odds	SE	Raw p	Rows / events
Confidential CbCR	exposure	exposure	0.951	1.447	0.511	61 / 31
Pooled supported cases	exposure	exposure	1.478	1.592	0.353	70 / 34
Confidential CbCR	Exposure and learning	Unstable	Withheld	Missing	-	52 / 27
Pooled supported cases	Exposure and learning	Unstable	Withheld	Missing	-	59 / 29

Original portfolio-exposure exercise, retained for comparison with the new JIBS-inspired tests. Standardized logit coefficients with country-clustered SE. Controls are unclassified portfolio weight and observed lagged market variables; family-year effects included. Learning uses pre-focal market dates; publication lags and historical quote coverage remain unaudited. Extreme learning slopes and invalid SE are withheld. The pooled sample covers audit oversight and confidential CbCR. CSVs retain odds ratios, Holm p-values, samples and failed-fit diagnostics.

Table 8 JIBS constructs and the implemented expansion

Mechanism	Koning et al. (2018)	Implemented measure and boundary
Competition	Adoption fraction among five countries with the most similar export patterns; EIU risk peers as alternative	COW raw exports: five highest Pearson correlations with at least 20 common observed destinations; fixed 2000, fixed 2013 from 2014, and annual-lag to2015. At least 3/5 dated policy peers (4/5 sensitivity). Unknown peers and bounds retained. EIU data unavailable.
Learning	Among ten geographic neighbours, median GDP growth of prior adopters exceeds nonadopters; GDPpc alternative	Ten GeoDist capital-distance neighbours, priorG<=t-2 versus laterG>t-1, signals datedt-1; GDP/GDPpc, liquidity changes, and foreign benchmarked contrasts excluding the focal country/cluster from controls. Require 80 percent signal coverage per relevant group.
Emulation	Berry cultural/administrative composite; IFAC professional-advocacy membership alternative	Official-language and legal-origin peer adoption fractions plus fixed 2000 generic IGO similarity. Partial separate proxies: religion/composite and historical IFAC advocacy remain missing.
Coercion	IMF credits and foreign aid relative to GDP; positive-resource indicator alternative	Lagged IMFcredit+SDR/GDP, net aid/GDP, sum and complete-case positive indicator. SDR is not verified conditional lending; aid and credit measures do not establish pressure for any policy.

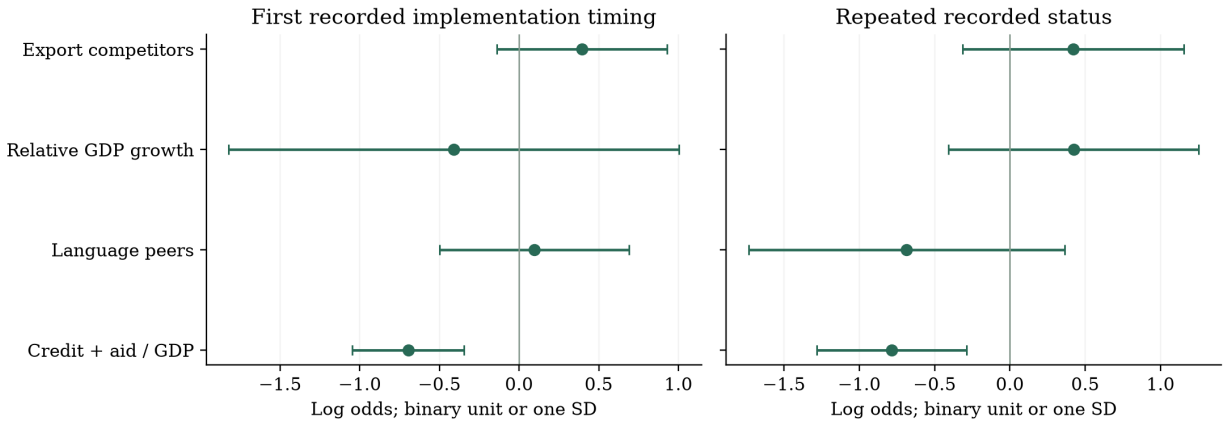
All focal features precede the focal calendar year. Contemporary data vintages and historical country/quote availability are not audited as real-time information sets. Geographic/language snapshots and legal-origin classifications are disclosed. The original JIBS main logit includes time effects without domestic controls; our single-mechanism baseline follows that choice and adds controlled and same-sample LPM sensitivities. The supplied JIBS paper remains the authority for its original design.

Table 9 Pooled JIBS-inspired mechanism associations

Design	Measure	Controls	Beta	SE	Raw p	All-test Holm p	Rows / countries
Timing	Export competitors	No	0.395	0.266	0.144	1.000	206/57
Timing	Export competitors	Yes	-0.051	0.328	0.877	1.000	191/54
Timing	Relative GDP growth	No	-0.409	0.692	0.558	1.000	72/31
Timing	Relative GDP growth	Yes	-0.754	0.859	0.388	1.000	67/29
Timing	Language peers	No	0.096	0.298	0.749	1.000	372/67
Timing	Language peers	Yes	-0.128	0.301	0.672	1.000	312/54
Timing	Credit + aid / GDP	No	-0.696	0.175	0.000	0.100	243/50
Timing	Credit + aid / GDP	Yes	-0.698	0.351	0.053	1.000	228/47
Status	Export competitors	No	0.420	0.367	0.257	1.000	516/57
Status	Export competitors	Yes	-0.262	0.388	0.503	1.000	488/54
Status	Relative GDP growth	No	0.423	0.416	0.313	1.000	256/63
Status	Relative GDP growth	Yes	0.264	0.431	0.544	1.000	233/59
Status	Language peers	No	-0.686	0.525	0.196	1.000	992/67
Status	Language peers	Yes	-0.867	0.612	0.162	1.000	877/55
Status	Credit + aid / GDP	No	-0.785	0.247	0.003	1.000	505/50
Status	Credit + aid / GDP	Yes	-0.774	0.384	0.050	1.000	472/47

Logit with family-year strata and country-cluster covariance; t reference countries-1df. Constant-outcome strata are removed. Continuous focal measures are standardized within the exact fitted sample; binary relative-growth learning remains 0/1. Controls: lagged log GDPpc, GDP growth and trade/GDP, plus peer-unknown coverage where relevant. Timing conditions on documented independent implementations; status is an absorbing recorded-date index, not a verified worldwide adoption census. Holm spans every 442 stable focal coefficient in the new logit/LPM matrix. Samples differ across rows; controlled changes cannot be attributed solely to controls. The full release includes exact row memberships, all families, combined models, LPM, benchmark sensitivities and failures.

Figure 6 Pooled mechanism associations and nominal uncertainty



Single-mechanism logit with family-year strata and coverage terms, without domestic economic controls. Bars are nominal 95 percent country-cluster/t intervals, not simultaneous intervals. Timing and recorded-status samples differ. No new coefficient passes the broad Holm correction; Table 9 and full CSV report multiplicity.

6 Identification and extension to the firm level

The follow-up completes the feasible weighting, factor, balanced-profile and mechanism analyses using observed local/public inputs. TableA 6 separates those completed tasks from missing data and legal validation. All source-paper tables and writing/method guidance remain in the reference JSON, now linked to the expanded methods and results. Clear definitions, table-specific sample explanations and restrained interpretation follow the reference papers; their prose is not copied.

The main unresolved identification work is at the issuer-policy boundary: first mandatory filing dates, fiscal alignment, eligibility and compliance, firm accounting/analyst controls, enforcement and voluntary disclosure. Incomplete country-family histories prevent a verified worldwide untreated census or an ordered legal-status model. The current static quote universe does not establish survival-free historical selection.

Historical IFAC advocacy, EIU risk peers, the full Berry composite, verified lending conditions and adoption-trigger series remain unavailable. The source archive retains 18 events without readable material,11 supported only by extracts and 51 flagged OCR pages for human review. These counts are not silently upgraded by the empirical extension. The original files, legal records and all source gaps remain with the project.

7 Conclusion

Disclosure regulations differ in information audience, eligible firms and implementation structures. Expanded liquidity comparisons reflect those differences, but economic weighting can also change their sign and magnitude. IFRS attenuation and the confidential-CbCR reversal show why the paper cannot infer a common market benefit from negative equal-country coefficients. Audit reporting provides a negative GDP-weighted descriptive comparison that warrants source- and eligibility-specific investigation.

The JIBS framework helps separate trade competition, foreign success, peer emulation and financial dependence. New proxies make substantially more of that framework measurable, while the full test matrix yields no coefficient that survives the broad multiplicity correction. A negative financial-dependence timing association survives only the narrower primary family. Neither result identifies a diffusion mechanism. Firm eligibility, complete legal histories and historically available information are the remaining requirements for linking capital-market effects to subsequent policy decisions.

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Appendix A Timing comparisons and remaining tests

Table A1 Log spread comparisons by calendar horizon

Family	g-2	g	g+1	g+2
IFRS	0.098	-0.156	-0.147	-0.074
ESG and CSR	-	-	-	-
Climate	-	-	-	-
ISSB	-0.023	-0.076	0.061	-
XBRL	-	-0.065	-	-
Governance	-0.029	0.056	0.030	-0.198
Beneficial ownership	0.021	-0.003	0.032	0.025
Confidential CbCR	0.086	0.006	-0.057	-0.034
Public CbCR	0.029	-0.187	-	-
Audit reporting	-0.067	0.050	0.025	-0.099
Audit oversight	-0.008	0.069	0.029	0.059
Extractive payments	-0.054	0.055	0.210	-
Supply chain	0.028	-0.026	0.066	0.053
Digital assets	-0.069	-0.030	0.016	0.042

Every column compares its indicated outcome year with g-1. Different columns can use different countries/cohorts and balanced securities; this is not a balanced event study. Negative values are relative improvements in log spread. A dash indicates insufficient support under the fixed rules. The g+1 column is primary; pre-period cells do not verify parallel trends.

Table A2 Test status and data requirements

Test	Current status	Required extension
Public versus confidential channel	Classified in policy register	Eligible firms, first public availability and policy-specific effects
Issuer and compliance heterogeneity	Design specified	Audited family-specific issuer and bindingness history
Formal and informal enforcement	Design specified	Predetermined measures relevant to each family
Voluntary firm disclosure	Not estimated	Guidance or comparable actual disclosure data
Clean cohort and event study	Balanced descriptive profiles completed	Verified treatment histories, balanced eligibility, anticipation and simultaneous inference
Adoption-trigger diagnostics	Not estimated	Family-relevant attention and incident series
Competition	Export-competitor tests completed	EIU risk peers unavailable; global histories incomplete
Learning	GDP/GDPpc and liquidity proxies tested	Historical information availability; timing liquidity samples remain sparse
Emulation	Language/legal/IGO proxies tested	Full cultural composite and historical IFAC advocacy unavailable
Coercion	Financial-dependence proxies tested	Verified policy-specific lending conditions unavailable
Ordinal adoption status	Not estimated	Comparable ordered legal status and proportional-odds diagnostics

Status describes this release. A planned or unavailable test is not reported as completed. Detailed source-paper methods, robustness checks and writing guidance are in reference-papers-and-expanded-study.json.

Table A3 Published ESG coding calendar year benchmark

Outcome	Change	Treated countries	Cohorts	Assignment cluster deletion range
Log spread	-0.016	27	12	-0.085 to 0.019
Log impact	-0.087	29	14	-0.148 to -0.040
Zero fraction	-0.006	29	14	-0.009 to -0.002

Source sample: supplied Krueger 2023 Table 1, PDF 49-50. Published 35 adopters and 30 never-coded countries, with calendar market outcomes restricted to2002-2020. France lacks the required baseline within this restricted market window. Balanced-security g+1 versus g-1 comparisons use later/never-coded controls,30 securities per country and 3 control countries per cohort. Known NFRD countries share an assignment cluster. The range is a deletion sensitivity, not a confidence interval. No full accounting controls, fiscal alignment or source analyst-screened sample are reproduced. The benchmark is separate from verified implementation estimates.

Appendix B Regulation source archive

The register and its source manifest link each event to preserved evidence without changing its inherited legal verification status. A saved source can be a legal instrument, an official explanatory page, a jurisdiction profile or author-released coding data. The latter remain published proxies. Original and linked sources are distinguished from partial text extracts and unavailable material.

Table A4 Regulation event JSON and source availability

Family	Event records	Saved source and text	Derivative or excerpt only	No readable source
IFRS	37	37	0	0
XBRL	4	4	0	0
Climate	3	2	0	1
ESG and CSR	41	40	0	1
Confidential CbCR	120	120	0	0
Governance	14	12	0	2
Beneficial ownership	24	17	1	6
Public CbCR	44	40	1	3
Audit reporting	38	34	2	2
Audit oversight	38	36	1	1
Extractive payments	10	8	0	2
Supply chain	11	8	3	0
Digital assets	33	30	3	0
ISSB	24	24	0	0

Source preservation audit on18 October 2026. The 441 records include 359 dated estimation cases and other event types; they are not 441 unique statutes. Saved source and text includes original or linked official documents and 39 author-repository-backed published proxies. Partial excerpts are not complete archived originals. No-readable-source includes inaccessible, unreadable and metadata-only captures. Counts are event-weighted: many events can refer to one source. All fourteen policy histories remain incomplete.

The local archive preserves 258 unique objects (187.4 MB), including originals and explicitly labelled derivatives. It records 4,593 examined PDF pages and 206 unique OCR page images. The 51 flagged OCR pages and the remaining low-text outputs require human review. A machine-readable gap list preserves the unresolved sources. The public companion distributes JSON metadata and original-source links; the document corpus, extraction logs and OCR derivatives are retained with the local study.

Appendix C Expanded-test coverage and completion audit

Table A5 GDP-weighted robustness coverage across expanded families

Configuration	Cells	Supported	Negative	Positive	Families
baseline	42	30	15	15	10
cap weighted quotes	42	30	13	17	10
median quote change	42	30	14	16	10
common three outcome sample	42	30	15	15	10
at least 200 days	42	27	12	15	9
exclude bottom cap quintile	42	30	14	16	10
unwinsorized proxies	42	30	17	13	10
five ninetyfive winsorized proxies	42	30	16	14	10
independent treated cases	42	30	15	15	10
independent cases both arms	42	24	12	12	8
exclude known other family events in window	42	24	11	13	8
exclude crisis windows 2008 2009 2020 2021	42	30	14	16	10
pretrend extrapolation adjustment	42	29	11	18	10
post years 1 to3 average	42	23	11	12	8

Counts refer to related descriptive cells across the original three outcomes and fourteen families; they are not independent experiments. The full grid retains each cell including failures. Configurations are applied separately, without searching combinations. PC1, equal-country counterparts, matched baselines and published ESG benchmark are in the released full matrix. Figure 5 shows GDP-weighted log-spread values for every family and configuration.

Table A6 Completion audit and unresolved inputs

Workstream	Current status	Evidence or remaining requirement
GDP and market-size weights	Completed	Six country weighting schemes; matched support, concentration and cluster deletions.
Factor, secondary outcomes and robustness	Completed	PC1, traded value/cap, volatility;14 configurations; balanced seven-year profiles; income split.
JIBS competition and learning	Implemented with disclosed coverage	Five export peers; geographic relative GDP/GDPpc growth; lagged liquidity and foreign benchmark signals. Sparse learning cells remain unsupported.
Emulation and financial dependence	Partial observable proxies completed	Language/legal/IGO and credit/aid measures. Full Berry composite, IFAC advocacy, EIU and verified loan conditions unavailable.
Firm-level causal design	Unfinished: required inputs missing	First mandatory filing dates, issuer eligibility, fiscal alignment, accounting/analyst controls, untreated histories and historical quote universe.
Policy-source verification	Unfinished: retained gap register	18 events lack readable source,11 excerpts only;51 OCR pages require review. Saving/OCR does not verify legal coding.
Other source-paper tests	Unfinished: required inputs missing	Policy-specific enforcement, voluntary guidance, triggers, IFAC histories and ordered legal status. No synthetic replacements.

This update completes feasible tests with the observed inputs and discloses all failures. It does not mark unavailable data, legal verification or source-review tasks complete. Original security/portfolio rows and source-document corpus remain local. All 441 expanded events and 51 inherited document codings remain in the source-linked JSON register.